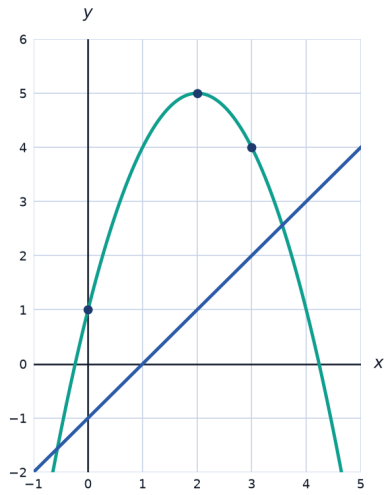
Quadratic functions

Core Concepts: Quadratic functions

1 For the model $f(x) = 2x + 5$ in quadratic functions, evaluate $f(2)$.



a. 9

b. 11

c. 4

d. 13

2 Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ and $x = 5$

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

3 A data set recorded for quadratic functions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 2x + 5$ in quadratic functions, evaluate $f(3)$.

a. 11

b. 13

c. 6

d. 15

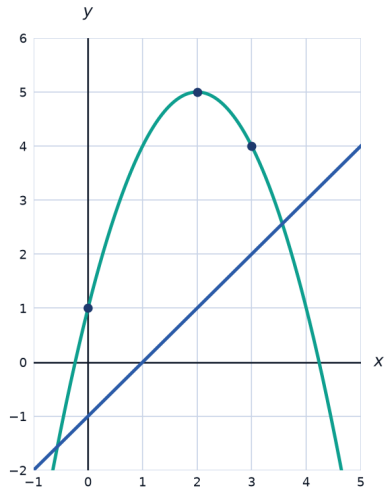
5 Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$



6 A data set recorded for quadratic functions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

Graphical & Geometric Interpretation

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7 For the model $f(x) = 2x + 5$ in quadratic functions, evaluate $f(4)$.

a. 13

b. 15

c. 8

d. 17

8 Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

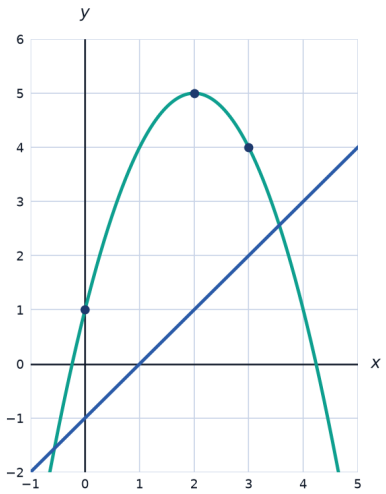
9 A data set recorded for quadratic functions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

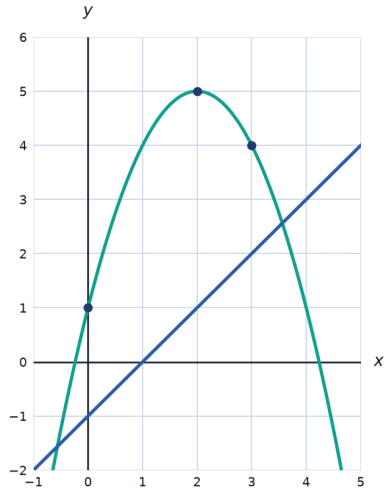
d. 26.5



Exam-Style Problem Solving & Modeling

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10 For the model $f(x) = 2x + 5$ in quadratic functions, evaluate $f(5)$.



a. 15

b. 17

c. 10

d. 19

11 Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ and $x = 8$

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

12 A data set recorded for quadratic functions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

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